

Path of Light: Interactive Narrative Design Based on Mix Reality for Silk Road Cultural Perception

Wenwen Yang*

Intelligent Media Design Lab
Communication University of China
Beijing, China
YangWenwen@cuc.edu.cn

Wanyi Miao*

Intelligent Media Design Lab
Communication University of China
Beijing, China
1184844287@qq.com

Guoyu Sun[†]

Intelligent Media Design Lab
Communication University of China
Beijing, China
gysun.sun@qq.com

Yeheng Wang

Intelligent Media Design Lab
Communication University of China
Beijing, China
2871902341@qq.com

Changran Zhao

Intelligent Media Design Lab
Communication University of China
Beijing, China
jasonzhao@cuc.edu.cn

Yilin Zhang

Intelligent Media Design Lab
Communication University of China
Beijing, China
yilinzhang0121@gmail.com

Wenxuan Kou

Intelligent Media Design Lab
Communication University of China
Beijing, China
playajoke@foxmail.com

Yaning Zhang

Intelligent Media Design Lab
Communication University of China
Beijing, China
202308233005@cuc.edu.cn

Xuran Nie

Intelligent Media Design Lab
Communication University of China
Beijing, China
1874023902@qq.com

Jingyu Liu

Intelligent Media Design Lab
Communication university of china
Beijing, China
1582855940@qq.com

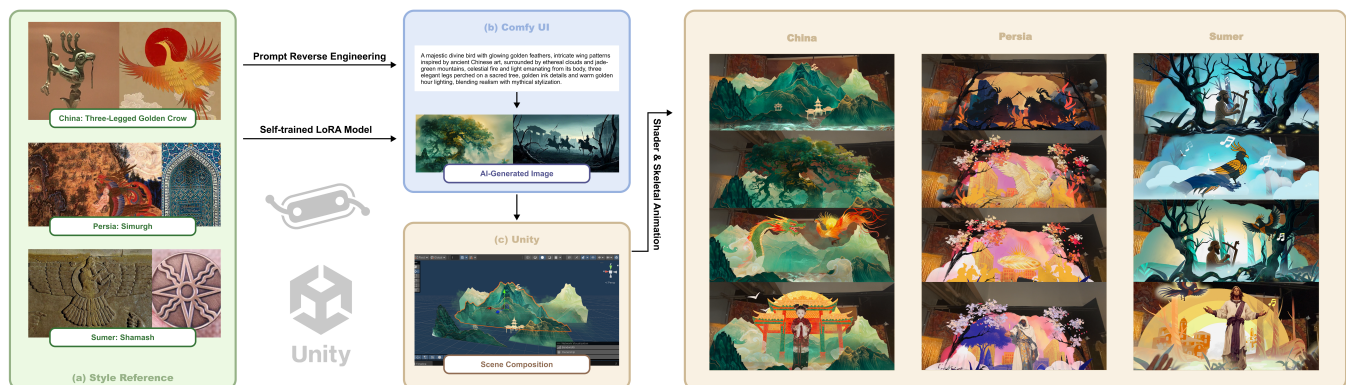


Figure 1: Mythical Scene Design Workflow.

*Both authors contributed equally to this research.

[†]Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '25, Busan, Republic of Korea

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758380>

Abstract

The Silk Road, spanning thousands of years, houses magnificent remnants of civilization. Among these, Silk Road mythology serves as a crucial component of intangible cultural heritage (ICH), crystallizing the primordial spirit and beliefs of diverse civilizations and offering a unique perspective for understanding a nation and its culture. However, these myths currently face challenges in preservation and dissemination. To address this, we have developed an immersive interactive system, *Path of Light*, that integrates artificial

intelligence (AI), mixed reality (MR), and natural interaction technologies, reinterpreting the mythical bird legends from three major ancient civilizations along the Silk Road. Within this system, participants embody “Light Seekers”, traversing mythological landscapes of different civilizations alongside the bird, personally experiencing these mythological narratives while exploring their underlying cultural significance. Feedback from the pilot study demonstrates that this interactive system stimulates audience interest in Silk Road mythology and provides an innovative cross-cultural communication model and perspective for disseminating ICH.

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → *Mixed / augmented reality*.

Keywords

Silk Road Culture, Mix Reality, Cross-cultural Communication, Embodied Cognition

ACM Reference Format:

Wenwen Yang, Wanyi Miao, Guoyu Sun, Yeheng Wang, Changran Zhao, Yilin Zhang, Wenxuan Kou, Yaning Zhang, Xuran Nie, and Jingyu Liu. 2025. Path of Light: Interactive Narrative Design Based on Mix Reality for Silk Road Cultural Perception. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758380>

1 Introduction

For thousands of years, the Silk Road served as a vital bridge for cultural exchange between Eastern and Western civilizations[4]. The mythological narratives along this route share similar structures[3] while adapting to local cultural contexts, forming the foundation for cross-cultural heritage dissemination[8]. However, these myths face significant preservation and popularization challenges: their abstract complexity hinders public comprehension, and traditional presentation methods fail to engage modern audiences, particularly younger generations.

Recent advances in AI and MR technologies have shown promise in preserving tangible cultural heritage[1, 5, 7, 10, 12, 13], yet their application remains limited in ICH of the oral traditions and expressions type[11], such as Silk Road mythology. To address these dissemination challenges, this study builds on previous research[6, 9] to propose an innovative immersive MR narrative method that integrates AI-generated content with mixed reality technologies. This approach transcends traditional VR/AR limitations by bringing Silk Road myths from digital screens into real-world spaces, facilitating more effective cultural transmission and encouraging greater youth engagement with intangible cultural heritage.

2 Design Process

Through an in-depth study of mythologies from countries along the Silk Road, a recurring motif was identified in which light is symbolically associated with mythical birds, often embedded within heroic narrative structures[2]. Therefore, we selected the Chinese Golden Crow, Persian Simurgh, and Sumerian Shamash myths as prototypes for story adaptation. To generate scene designs that ensure

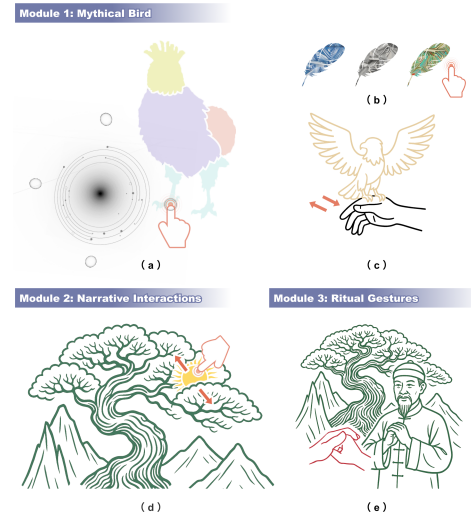


Figure 2: Embodied Interaction Design.

cultural authenticity, this project constructed a highly controllable workflow for scene asset generation, as shown in Figures 1(a) and (b). First, we collected relevant mythological illustrations and historical materials reflecting local artistic styles, then conducted LoRA fine-tuning based on these materials to build generative models capable of reproducing the artistic characteristics of various Silk Road regions, while simultaneously extracting optimal prompts through reverse prompt inference from mythological images. Subsequently, the LoRA models and prompts were integrated into the ComfyUI platform to establish a complete automated generation workflow. Leveraging our collective design expertise, we conducted dual assessments of aesthetic quality and cultural authenticity on all generated results, selecting materials that met our standards and performing inpainting and manual corrections. Finally, the scene assets were enhanced with dynamic effects through shaders and skeletal animations in Unity, and assembled into complete story scenes, as shown in Figure 1(c).

3 System Prototype

The system of *Path of Light* was developed using the Unity 6000.0.23f1 and Vision Pro to construct 2D animated narratives displayed within a three-dimensional spatial context. By leveraging the spatial computing capabilities of the Vision Pro platform, the project realized precise localization and realistic presentation of content in the physical environment through spatial anchors and high-performance rendering optimization techniques. In the system, the storylines from the three countries are presented in a parallel structure. Each storyline consists of three interactive experience modules: Mythical Bird, Narrative Interactions, and Ritual Gestures, as shown in Fig. 2.

3.1 Module 1. Mythical Bird

This module consists of two main phases: constructing the mythical bird and engaging in natural interaction with the bird. In the construction phase, viewers can assemble the shape of the mythical

bird by tapping four directional indicators (up, down, left, right) on the star chart, as shown in Fig. 2(a). Once the bird's shape is complete, users can select one of three pre-designed AIGC-generated stylized feather patterns to color the bird, as depicted in Fig. 2(b). Upon completing the texturing process, the bird flaps its wings and flies off toward the viewer's field of vision, gradually disappearing. At the point of its disappearance, a portal opens, symbolizing the entrance into a foreign cultural context and initiating Module 2: Narrative Interactions. After completing modules 2 and 3, viewers can naturally interact with the mythical bird. When a viewer extends their hand, the soaring bird hovers and lands gently upon it, as shown in Fig. 2(c); when the hand is withdrawn, the bird flies off into the distance.

3.2 Module 2. Narrative Interactions

Upon entering the portal, local natural scenery gradually emerge around the viewer. Simultaneously, an audio narration of the relevant myth begins to play. As the buildings fade away, a myth narrative scene comes into view. Each myth features four sequential scenes. Viewers follow the cues to touch specific objects within each scene, triggering animations and plot developments to unlock the subsequent scene, as shown in Fig. 2(d).

3.3 Module 3. Ritual Gestures

In the final scene of the narrative interaction, the viewer encounters a character dressed in traditional local attire. Following cues in the scene, the viewer is required to perform a culturally inspired gesture of greeting to express goodwill. This project independently developed a gesture recognition system based on XR Hand data. Once the system successfully recognizes the viewer's greeting gesture, the character reciprocates with the same motion, as illustrated in Fig. 2(e).

4 Conclusion and Future Work

Drawing upon the myths and legends of the Silk Road, this study designed and developed an immersive MR system called *Path of Light*. A pilot study was conducted with 10 participants (aged 19-52; three male, seven female). All participants expressed highly positive feedback about the system, with three individuals noting a sense of cultural resonance, stating, "It was surprising to discover how similar the Persian legend of the divine bird is to those from our own country." Preliminary interview results indicate that the system's narrative and interaction design strategies effectively facilitated participants' deepened understanding of Silk Road culture, sparked their interest, and enhanced their cross-cultural awareness. In the future, we plan to include routes from additional countries and conduct larger-scale experiments to validate the system's practical impact rigorously.

Acknowledgments

Supported by The National Social Science Fund of China (Grant No. 24BG143)

References

- [1] Federico Becattini, Pietro Bongini, Luana Bulla, Alberto Del Bimbo, Ludovica Marinucci, Misael Mongiovi, and Valentina Presutti. 2023. VISCOUNT: A Large-scale Multilingual Visual Question Answering Dataset for Cultural Heritage. *ACM Trans. Multimedia Comput. Commun. Appl.* 19, 6, Article 193 (July 2023), 20 pages. doi:10.1145/3590773
- [2] Christopher I. Beckwith. 2009. *Empires of the Silk Road: A History of Central Eurasia from the Bronze Age to the Present*. Princeton University Press, Princeton.
- [3] Joseph Campbell. 1949. *The Hero with a Thousand Faces*. Pantheon Books, New York.
- [4] Peter Frankopan. 2016. *The Silk Roads: A New History of the World*. Bloomsbury, London.
- [5] Carola Gatto, Maria Cristina Barba, Sofia Chiarello, Laura Corchia, Federica Faggiano, Benito Luigi Nuzzo, Ileana Riera Panaro, Giada Sumerano, Valerio De Luca, Manuela De Giorgi, and Lucio Tommaso De Paolis. 2025. Improving accessibility to cultural heritage: Integration of extended reality, tactile prints and user experience analysis for the church of Madonna dell'Itri. *J. Comput. Cult. Herit.* (May 2025). doi:10.1145/3733154 Just Accepted.
- [6] Wenchen Guo, Wenbo Zhao, Guoyu Sun, Yanni Li, Yangyi Ye, and Su Wang. 2024. Enhancing the Digital Inheritance and Embodied Experience of Zen based on Multimodal Mixed Reality System. In *ACM SIGGRAPH 2024 Posters* (Denver, CO, USA) (SIGGRAPH '24). Association for Computing Machinery, New York, NY, USA, Article 12, 2 pages. doi:10.1145/3641234.3671076
- [7] Xi He and Xing Sun. 2024. Research on the Reconstruction of Ming Dynasty History Based on AIGC. In *Proceedings of the Eleventh International Symposium of Chinese CHI* (Denpasar, Bali, Indonesia) (CHCHI '23). Association for Computing Machinery, New York, NY, USA, 449–454. doi:10.1145/3629606.3629652
- [8] Johann Gottfried Herder. 1989. *Ideen zur Philosophie der Geschichte der Menschheit*. Deutscher Klassiker Verlag, Frankfurt am Main.
- [9] Jingke Li, Guoyu Sun, Congyun Tang, Wenjuan Chen, Wenwen Yang, Wenxuan Kou, Zhonghe Ruan, Wanqing Ma, and Xuran Nie. 2024. Silk Road Journey: A Real-time AI-based Interactive Art Installation for Silk Road Cultural Reenactment and Experience. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 418, 5 pages. doi:10.1145/3613905.3648655
- [10] Kexin Nie and Mengyao Guo. 2024. Flying Your Imagination: Integrating AI in VR for Kite Heritage. In *SIGGRAPH Asia 2024 Posters* (SA '24). Association for Computing Machinery, New York, NY, USA, Article 30, 2 pages. doi:10.1145/3681756.3697907
- [11] Qian Tan, Khairul Manami Kamarudin, and Siti Sarah Herman. 2025. Systematic Review of Empowering Intangible Cultural Heritage with Metaverse Technology. *J. Comput. Cult. Herit.* 18, 2, Article 26 (May 2025), 18 pages. doi:10.1145/3723169
- [12] Yingna Wang, Qingqin Liu, Xiaoying Wei, and Mingming Fan. 2025. Facilitating Daily Practice in Intangible Cultural Heritage through Virtual Reality: A Case Study of Traditional Chinese Flower Arrangement. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 1126, 17 pages. doi:10.1145/3706598.3713409
- [13] Jian Yu, Zhan Wang, Yifan Cao, Hao Cui, and Wei Zeng. 2025. Centennial Drama Reimagined: An Immersive Experience of Intangible Cultural Heritage through Contextual Storytelling in Virtual Reality. *J. Comput. Cult. Herit.* 18, 1, Article 11 (Feb. 2025), 22 pages. doi:10.1145/3705613